Lecture:4 - Addressing Modes

The different ways in which a microprocessor can access data from memory locations are referred to as its addressing modes.

MAP EVERYTHING/CALCULATE [SEGMENT:OFFSET] OF ANY INSTRUCTION FOLLOWING THIS TABLE:

Segment		Offset Registers	Function
CS		IP	Address of the next instruction
DS		BX, DI, SI	Address of data
SS	-	SP, BP	Addresses in the stack
ES		BX, DI, SI	Address of destination data (for string instructions)

Addressing modes and categories are mainly of 5 types:

1) Addressing Data
2) Addressing Program codes in memory
3) Addressing Stack in memory
4) Addressing I/O
5) Implied addressing

- Instruction format: | opcode | Operand(s) |
- Instructions can have 1, 2 or no operands
  - **INC AX** ; 1 operand
  - **ADD CX, DX** ; 2 operands ==> CX = CX + DX
  - **HLT** ; no operand
- Instruction cannot have:
  - **SUB [DI], [1234h]** ; memory locations as both operands
  - **MOV 1234, AX** ; immediate data as destination operand

MOD / R/M	00	01	10	11 W = 0	11 W = 1
000	[BX] + [SI]	[BX] + [SI] + d8	[BX] + [SI] + d16	AL	AX
001	[BX] + [DI]	[BX] + [DI] + d8	[BX] + [DI] + d16	CL	CX
010	[BP] + [SI]	[BP] + [SI] + d8	[BP] + [SI] + d16	DL	DX
011	[BP] + [DI]	[BP] + [DI] + d8	[BP] + [DI] + d16	BL	BX
100	[SI]	[SI] + d8	[SI]+d16	AH	SP
101	[DI]	[DI] + d8	[DI]+d16	CH	BP
110	d16 (direct address)	[BP] + d8	[BP]+d16	CH	SI
111	[BX]	[BX] + d8	[BX] + d16	BH	DI

1) Addressing Data

To calculate the Physical address of these instructions:
DS : BX/ DI/ SI [Following the table]
PA = DS * 10h + offset; // here the offsets will be either **BX/DI/SI**

Addressing data again has 7 sub classifications:

1. Immediate addressing
2. Direct addressing
3. Register [direct] addressing
4. Register indirect addressing
5. Base-plus-index addressing
6. Register relative addressing
7. Base-relative-plus-index addressing

Examples of each:

1.Immediate addressing:
MOV AX, **12h**; // a direct hex value in source

2.Direct addressing:
MOV BX, [1234H]; // An offset is given in source/destination and other one is a register

3.Register [direct] addressing:
MOV AX,BX ; // both are registers

4.Register indirect addressing:
MOV CX, [BX] ; // Offset value is provided in a register in **BX/DI/SI** either is source or destination

#Exception:
MOV CX, [BP] ; // if BP is in the offset then it will be considered as register relative addressing since a displacement of 00h is automatically added after it like [BP+00H] and to find out the Physical address of in the memory we will add the BP offset with SS segment's value instead of DS. => **SS * 10h +[BP]#**

5. Base-plus-index addressing:
MOV DX, [BX+DI] ; // Base and Index **[BX/DI/SI]** are given in offset part either is source or destination

6.Register relative addressing:
MOV AX, [BX+1000h] ; // A displacement is added with the offset either in the source or destination

7.Base-relative-plus-index addressing:
MOV AX, [BX+DI+12h] ; // A base, index and displacement will be in the offset either in the source or destination

Example:

Given, BX = 0300H, SI = 0200H, DS = 1000H, ARRAY = 1000H

Instruction	Addressing mode	Source	Destination
MOV AX, BX	Register Direct Addressing	BX = 0300H	AX
MOV AH, 3AH	Immediate Addressing	3AH	AH
MOV [1234H], AX	Direct Addressing	AX	11234H
MOV [BX], CL	Register Indirect Addressing	CL	10300H
MOV [BX+SI],DL	Base-plus-index Addressing	DL	10500H
MOV CL, [BX + 4H]	Register Relative Addressing	10304H PA = 1000 * 10H + (0300 +4) = 10304H	CL
MOV ARRAY[BX +SI], DX	Base-relative-plus-index Addressing	DX	11500H DS*10H + (0300+0200+ 1000) = 11500H

2) Addressing Program Codes

****Used with JMP or CALL instructions usually.**

Addressing codes again has 3 sub-classifications:
i) Direct
ii) Indirect
iii) Relative

To calculate the Physical address of these instructions:
CS : IP [Following the table]
PA = CS * 10h + IP;
// here the offset/ IP will vary:

- If IP is given in BX/DI/SI registers **as offset** then first we have to find out DS*10h + [BX/DI/SI] (this is the physical address where IP is located) first then add this IP with CS*10h
- If IP is given in BP register as offset the first we have to find out SS*10h + [BP]. (this is the physical address where IP is located) then add this with CS*10h

[Check examples for better understanding]

1.Direct:
JMP BX ; // any general purpose register can be here AX/BX/CX/DX or SP/BP/DI/SI, **here, AX/BX/CX/DX/SP/BP/DI/SI**

2. Indirect:
JMP [BX] ; // **[BP]./[BX].[DI].[SI]**

3. Relative:
JMP [BX + 100h] ; // any relative register with displacement

Examples:

Address	10600h	10601h	20600h	200ABh	200ACh	33412h	39A87h	39A88h
Data	12h	34h	56h	87h	9Ah	B0	FEh	20h

A portion of a memory is given above, where there are memory locations and their corresponding instruction bytes or data. Here, DS = 2000h, SS = 1000h, CS = 3000h, BP = 0400h, SI = 0200h, BX = 00AB.

a) Find out what will be the physical address of the memory for **CALL [BX]** instruction. Also mention which data we will get from this location from the given table.

b) Find out what will be the physical address of the memory for **JMP[BP + SI]** instruction. Also mention which data we will get from this location from the given table.

Solution:
a)
As **Call [BX]** is an **addressing codes in program [indirect]** type addressing mode.
The physical address will be = CS * 10h + IP

Now, the value of IP is in the given table at the physical address = (DS *10 +BX)
= (2000*10 + 00AB)
= 200ABh

Therefore, the IP is at 200ABh of the table. IP = 9A87h

So, the physical address of CALL [BX] is = CS *10h + IP
= 3000h * 10h + 9A87h
= 39A87h;
The data that we will get from this location/ after executing the instruction is = FEh.

b)
As **JMP[BP + SI]** is an **addressing codes in program [Relative]** type addressing mode.
The physical address will be = CS * 10h + IP

Now, the value of IP is in the given table at the physical address = (SS *10 +[BP+SI])
= (1000*10 + 0400 + 0200)
= 10600h

Therefore, the IP is at 200ABh of the table. IP = 3412h.

So, the physical address of JMP[BP + SI] is = CS *10h + IP
= 3000h * 10h + 3412h
= 33412h;
The data that we will get from this location/ after executing the instruction is = B0h.

3) Addressing Stack

****Push/ POP/ CALL instructions are usually associated with it.**
PUSH AX ;
POP CX ;
CALL SUM; // SUM is a procedure name

To calculate the Physical address of **Stack for these instructions**:
SS : SP [Following the table]
PA = SS * 10h + SP;

4. Addressing Input and Output Port

*IN and OUT instructions are used to address I/O ports
i) direct addressing

IN AL, 05h ; Here 05h is a input port number

ii) Indirect addressing

OUT DX, AL ; DX contains the address of I/O port

*Only the DX register can be used to point at an I/O port otherwise not.

5. Implied Addressing

*No explicit address is given with the instruction
*Implied within the instruction itself
Examples:

CLC ; clear carry flag
HLT ; halts the program
RET ; return